Sizing and Configuration Decisions

Evidence, options and recommendations for the twelve open sizing and configuration decisions, ahead of the configuration lock on 17 October and the numbers freeze on 20 October 2026.

Decisions in this brief

D1	Wing area and planform	D7	Tail
D2	Accepted stall speed	D8	Landing gear
D3	Payload unit and sensor	D9	Propulsion layout
D4	Mission 1 waiver	D10	Sensor-bay closure
D5	Drop height	D11	Body shape
D6	Wing position	D12	D&R architecture

Document	Revision	Prepared by	Institution
AUM-DBF-27-DB-001	A, 14 October 2026	Badr Obtel	American University of the Middle East

01

Summary

Purpose

1.1

The configuration locks on 17 October and the design numbers freeze on 20 October. This brief brings the evidence for each open sizing and configuration decision into one place. For every decision it sets out the options, the supporting numbers and the risks, then gives a recommended option and the finding that would overturn it. Table 01 doubles as the team's decision record (Section 8).

Recommended baseline

1.2

The analysis recommends a straight, flapped wing of 0.9 m² carrying a 4 kg payload unit, sized for a 14 m/s stall at Mission 2 weight, with Mission 1 waived and a closed-loop drum for sensor deployment and recovery. Across 120 draws of the uncertain inputs this design averages 6.33 of the 7.5 available mission points, and Mission 2 stays feasible in 98 % of draws.

Basis

1.3

The figures come from analysis Scripts 1 to 6 and the Table 06 deployment-and-recovery (D&R) trade. The model carries the D&R mechanism on every mission, as rule §3.1.2 requires. It is calibrated on 452 values verified against the 2021–26 top-3 AIAA design reports and on 513 rows of published fly-off results.

Decisions open	First deadline	Wing area	Payload unit	Mean total score	Mission 2 feasible
12	17 Oct	0.9 m²	4 kg	6.33 of 7.5	98 %

02

Method

Figure of merit

2.1

Designs are ranked by total mission score, $M_1 + M_2 + M_3 + GM$, which is capped at 7.5. Each mission is scored against the expected field, defined as 75 % of the per-mission specialist maxima anchored on the 2021–26 AIAA reports.

Uncertainty

2.2

Every design is evaluated on the same 120 random draws of the uncertain inputs (common random numbers), so any two designs are compared draw by draw. Three inputs are resampled from measured data:

- lap-time realism, from 24 flight pairs (median ×1.15);
- pack specific energy, from 9 packs;
- airframe mass, from the fit scatter (±31 % at one standard deviation).

Drag, efficiency, $C_{L,\max}$, air density and container mass are drawn from estimated ranges. “Mean” denotes the average over the draws, and P10 and P90 mark the boundaries of the worst and best tenth.

Evidence grades

2.3

Each input carries one of three grades:

- *verified*: quoted from a primary source and machine-checked;
- *reseller*: taken from a retail listing only;
- *estimate*: a reasoned range, sampled in the Monte Carlo.

Model limits

2.4

The model assigns no cost to take-off and landing risk, structural failure, build schedule or pilot skill. These factors enter through the risk register (Section 7) and the team’s judgment.

03 Decision summary

Table 01 lists every decision with its recommended option, the confidence attached to that recommendation and the finding that would change it. Confidence grades the strength of the supporting evidence only. The final column is left blank for the team to record its choice.

Table 01

Open decisions and decision record. Recommendations come from the analysis; the team records its decision in the shaded column (Section 8).

ID	Decision	Recommendation	Confidence	Changes if	Team decision
D1	Wing area, planform	Straight, flapped, 0.9 m² (0.9–1.0)	High	Stall below 13 m/s (then 1.0–1.3 m²)	
D2	Accepted 1 g stall at M2 weight	14 m/s (167 N/m²)	Pilot's call	Pilot's number; +0.13–0.15 per m/s above 14	
D3	Payload unit and sensor	4 kg unit (3.4 kg sensor), simulators as mass allows; sensor declared at tech	High	First weigh-in far below the fit	
D4	Mission 1	Waive (credited by M2)	High	A Q&A ruling on the credit	
D5	Drop height	Design 1.524 m; declare after TC-02	Medium	TC-02 fails at 1.524 m	
D6	Wing position	High wing	Medium	CAD favours a low-wing exit or loading path	
D7	Tail	Conventional	Medium	CAD clearance check fails	
D8	Landing gear	Tricycle, fixed, faired	Medium	Pilot prefers a taildragger	
D9	Propulsion layout	Single tractor	Low (pre-study)	The propulsion study (next)	
D10	Sensor-bay closure	Door, unless an open slot costs under 2.4–4.1 cm²	Low	WP-01 CFD on the slot	
D11	Body shape	From packaging: two units, reel, sensor bay	Deferred	TC-02 sets the container	
D12	D&R architecture	A, closed-loop drum	Medium-high	TC-05 bench counts	

Airfoil

3.1

The airfoil stays as selected in the proposal. Its Fig. 05 XFoil study across 13 sections chose the SD7037 for the race envelope. The team's airfoil matrix is retired because its own arithmetic reverses its winner.

04 Sizing

Wing area and planform

4.1 Decision D1

The robust design map (Fig. 01) scores every combination of wing area and payload unit over the 120 draws. The best point is a 0.9 m² wing with a 4 kg unit: a mean of 6.33, with Mission 2 feasible in 98 % of draws. Near-optimal designs form a plateau from 0.8 to 1.0 m². On the same draws, with each candidate at its best unit, the alternatives score as follows:

- 1.0 m²: 6.29;
- 0.8 m²: 6.23, with 1.0 m² ahead in 65 % of draws;
- 0.6 m²: 5.78;
- the 0.414 m² placeholder: 5.09;
- the delta: 4.60.

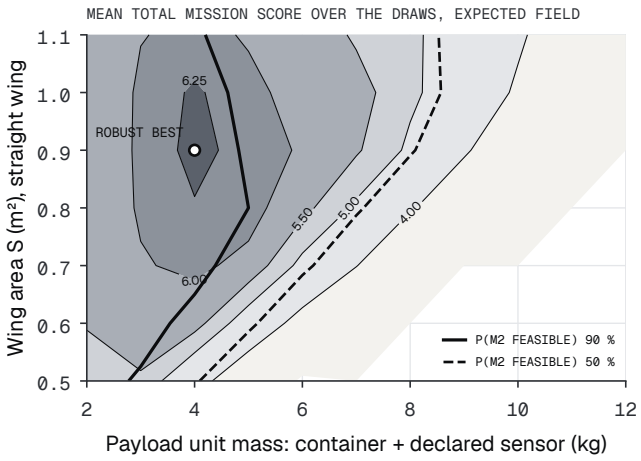


Figure 01

Mean total mission score over 120 draws peaks at 0.9 m² with a 4 kg unit. Straight wing, 14 m/s stall, reel included. Solid and dashed lines mark P(M2 feasible) of 90 % and 50 %. Source: `s6_report.txt`.

The straight wing outscores both the placeholder and the delta in 99 % of draws.

The lower edge of the plateau carries a specific penalty. With the reel's 0.32 kg on board, a wing 10 % smaller loses the payload step, which costs 0.24 points in the tornado study. The wing should therefore be sized at 0.9 m² or slightly above.

The planform is straight and flapped, and the delta remains only as a comparison case. Taper ratio will be set from XFLR5 results (span efficiency and stall progression) together with manufacturing constraints. The team's planform matrix is retired because four of its five criteria were copied from the wing-position matrix.

Recommendation

A straight, flapped wing of 0.9 m², within a plateau of 0.9–1.0 m².

Accepted stall speed

4.2 Decision D2

The stall bound fixes the Mission 2 wing loading through $W/S = \rho V_s^2 C_{L,\max}/2$. At 14 m/s, with $\rho = 1.064$ kg/m³ and $C_{L,\max} = 1.6$, it gives 167 N/m². This is the largest single lever in the model (Fig. 02, Table 02): each 1 m/s above 14 m/s adds 0.13–0.15 to the mean score, and lowering the bound to 11 m/s costs 1.4.

The model applies no take-off distance limit and assigns no cost to landing risk, so this gain carries risk it leaves unpriced. The 2025 fly-off shows what is at stake.[1]

- The two heavy top-3 designs, UNSW and ERAU, flew at 248–257 N/m² (about 17 m/s) and both scored zero in Mission 2.
- UCLA, at about 169 N/m², scored in every mission.

Recommendation

14 m/s. Raise the bound to 15 m/s (+0.13) only if the pilot signs off after seeing the landing speed.

Payload unit and sensor

4.3 Decision D3

Simulators must match the container plus the declared sensor to within ±28.35 g (§3.1.3), so Mission 2 payload comes in whole units. The declared sensor mass also sets the ground-mission score. Table 03 shows the trade at 0.9 m².

[1] Wing loadings derived from the quoted Mission 2 gross mass and wing area of each report (`analysis/data/dbf_aircraft.toml`). UCLA: 18.73 lb on a 6 ft span with a 10.59 in mean chord (0.49 m²). Fly-off scores from `dbf_results.csv`. The 17 m/s equivalent uses this brief's ρ and $C_{L,\max}$.

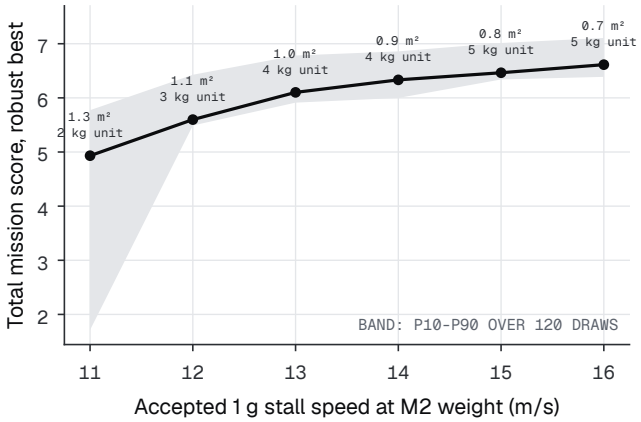


Figure 02

Each extra 1 m/s of accepted stall adds 0.13–0.15 to the robust best design's mean. Wing area grid 0.6–1.4 m², units 2–9 kg. Band: P10–P90 over the 120 draws. Source: `db01_report.txt`.

Table 02

Stall-bound sweep. W/S at $C_{L,max}$ 1.6 and ρ 1.064 kg/m³. Source: `db01_report.txt`.

Stall, m/s	W/S, N/m²	Wing, m²	Unit, kg	Mean	P(M2)
11	103	1.3	2	4.93	86 %
12	123	1.1	3	5.60	91 %
13	144	1.0	4	6.10	93 %
14	**167**	**0.9**	**4**	**6.33**	**98 %**
15	192	0.8	5	6.46	95 %
16	218	0.7	5	6.61	98 %

Table 03

The 4 kg unit leads on both mean and P10 at 0.9 m² and 14 m/s. Sensor mass is the unit less the 0.6 kg container; mission columns are means. Source: `db01_report.txt`.

Unit, kg	Mean	P10	P(M2)	M2	M3	GM
2	5.62	5.25	99 %	1.48	2.43	0.72
3	6.07	5.61	99 %	1.47	2.72	0.88
4	**6.33**	**6.00**	**98 %**	**1.43**	**2.85**	**1.05**
5	6.15	2.25	88 %	1.29	2.64	1.22
6	5.96	2.37	80 %	1.18	2.40	1.38
8	4.89	2.50	52 %	0.84	1.55	1.50

The 4 kg unit gives both the best mean and the best P10. Heavier units raise the median and the ground-mission score, and an 8 kg unit reaches the GM ceiling of 1.50. They also make Mission 2 infeasible whenever the airframe comes in heavy. P(M2) falls from 98 % at 4 kg to 52 % at 8 kg, and P10 collapses to the floor set by Mission 1 and the ground mission alone. A failed Mission 2 also blocks Mission 3.

The declared sensor mass is fixed only at technical inspection in April (§4.1), which keeps the choice open. The payload bay should be sized now for two container-sized units, with the sensor mass chosen after the first weigh-in and flight tests.

Recommendation

A 4 kg unit carrying a 3.4 kg sensor, with simulators added as mass allows. Declare the sensor at technical inspection.

Mission 1 waiver

4.4 Decision D4

A successful Mission 2 also credits Mission 1, and flight attempts are scarce: teams that passed technical inspection received 2.4–2.9 official attempts each from 2022 to 2026. At the field's 65 % success rate per attempt, with three attempts, waiving Mission 1 raises the probability of completing Mission 3 from 40 % to 72 %. With four attempts the rise is from 70 % to 87 %. Every Script 6 number in this brief assumes the waiver.

The waiver gives up one fallback: if Mission 2 fails, no Mission 1 score remains to fall back on.

Recommendation

Waive Mission 1 and take its credit through Mission 2.

Drop height

4.5 Decision D5

The ground-mission score rises linearly with the declared drop height, up to 1.524 m (60 in). The container should be designed for 1.524 m. The team should then declare the greatest height that TC-02 passes cleanly with the real sensor, before the design report.

Recommendation

Design for 1.524 m and declare the height after TC-02.

05

Configuration

Field practice

5.1

Field practice leaves the configuration open. Of the 19 verified top-3 aircraft since 2021, 13 flew a high wing, 13 a tricycle gear, 16 a conventional tail and 12 a single motor (Fig. 03). No feature predicts fly-off success under Fisher's exact test:

- low wings completed Mission 3 in 5 of 6 cases, against 4 of 9 for high wings ($p = 0.29$);
- single motors completed it in 7 of 10 cases, against 1 of 4 for twins ($p = 0.24$);
- gear type shows no relationship at all ($p = 1.00$).

Each configuration choice therefore rests on physics, packaging and risk. The team's trade matrices were audited for arithmetic and evidence, then re-weighted with the Script 6 sensitivities. Table 04 gives the probability that each option ranks first over 5,000 perturbations.

	WING	GEAR	TAIL	MOTORS	
DSCE 21 –	HIGH	TRICYCLE	CONV.	1	NO FLY-OFF DATA
ERAU 21 –	HIGH	TRICYCLE	INV. V	2	NO FLY-OFF DATA
UCF 21 –	HIGH	TAIL-DR.	CONV.	1	NO FLY-OFF DATA
MICHIGAN 21 –	HIGH	TRICYCLE	CONV.	2	NO FLY-OFF DATA
JOANNEUM 22 –	HIGH	TRICYCLE	CONV.	1	● M3 DONE #3
MIT 22 –	HIGH	QUAD	T-TAIL	2	✕ M3 FAILED #32
WASHU 22 –	LOW	TAIL-DR.	CONV.	1	● M3 DONE #11
ERAU 23 –	LOW	TRICYCLE	CONV.	1	● M3 DONE #3
UW 23 –	LOW	TAIL-DR.	CONV.	1	● M3 DONE #5
VT 23 –	HIGH	TRICYCLE	CONV.	1	● M3 DONE #4
LJUBLJANA 24 –	HIGH	TRICYCLE	T-TAIL	1	✕ M3 FAILED #56
USC 24 –	HIGH	TRICYCLE	CONV.	2	✕ M3 FAILED #32
VT 24 –	HIGH	TRICYCLE	CONV.	2	● M3 DONE #4
ERAU 25 –	LOW	TRICYCLE	CONV.	1	✕ M3 FAILED #57
UCLA 25 –	HIGH	TAIL-DR.	CONV.	1	● M3 DONE #10
UNSW 25 –	HIGH	TRICYCLE	CONV.	2	✕ M3 FAILED #52
GT 26 –	LOW	TRICYCLE	CONV.	N/S	● M3 DONE #7
LJUBLJANA 26 –	LOW	TRICYCLE	CONV.	1	● M3 DONE #1
USC 26 –	HIGH	TAIL-DR.	CONV.	1	✕ M3 FAILED #21

Figure 03

No configuration feature of the 19 verified top-3 aircraft (2021–26) predicts its fly-off outcome. Source: `data/dbf_config.csv`.

Table 04

After the audit and Script 6 weighting, the high wing, conventional tail and tricycle gear each rank first in most perturbations. Source: `trades_report.txt`.

Matrix	Probability of ranking first
Wing position	High 95 %, low 5 %, mid 0 %
Tail	Conventional 78 %, T-tail 14 %, twin-boom 8 %
Landing gear	Tricycle 76 %, taildragger 24 %
Fuselage structure	Box-beam 100 % (manufacturing plan)
Airfoil, planform	Retired (arithmetic errors, copied criteria)

Wing position

5.2 Decision D6

A high wing wins 95 % of the audited trade. The audit corrected the teammates' low-wing payload score, because four low-wing top-3 designs (WashU 2022, UW 2023, and ERAU in 2023 and 2025) chose the low wing specifically for payload access. Three factors settle the final call: the sensor exit path, staging access through a top or bottom hatch (a Q&A on the §3.3.2 hatch rule is pending) and the pilot's preference.

Recommendation

High wing, at medium confidence.

Tail

5.3 Decision D7

The as-made matrix chose a T-tail on line clearance. The tow line hangs 31–86° below horizontal for sensors of 0.5–5 kg, which makes clearance a function of where the line exits the fuselage, with the tail type playing little part. A T-tail's fin also carries the stabiliser loads, which makes it heavier and more prone to flutter. The clearance should be confirmed in CAD once the exit is placed.

Recommendation

Conventional tail, at medium confidence.

Landing gear

5.4 Decision D8

A retractable system pays for itself only below 120–240 g. That break-even is the fixed gear's drag area (9.9 cm² measured on Georgia Tech 2026; 19.4 cm² predicted for FH Joanneum 2022) multiplied by the exchange rate of 12.3 g per cm². Retracting would save at most 2.6–5.1 s of Mission 2 time, about 0.01 points, against the added risk of a gear-up landing. A fairing should reduce the fixed gear's drag further; its benefit has yet to be quantified from a verified source.

Recommendation

Fixed, faired tricycle gear, at medium confidence.

Propulsion layout

5.5 Decision D9

A nose-mounted tractor keeps the propeller disc clear of the tow line, whose slack moments during deployment and stowage are the SR-10 risk. The field also favours single motors (12 of 19), and twins completed Mission 3 in only 1 of 4 cases. The coming propulsion study (APC and UIUC data, a motor model, 12S against 8S) will confirm or overturn this choice.

Recommendation

Single tractor, provisional until the propulsion study.

Sensor-bay closure

5.6 Decision D10

Rule §3.1.1 requires an opening left uncovered in Mission 3 to stay uncovered in Missions 1 and 2. At the exchange rate of 10 cm² of drag area per 123 g, a door mechanism of an estimated 30–50 g pays for itself if the open slot's drag area exceeds

2.4–4.1 cm². A slot sized for a can-sized dart with fins probably exceeds that threshold, though no verified cavity-drag figure exists; WP-01 CFD should compute it. A door adds one failure mode, a door that stays shut and prevents deployment, which the latch test covers.

Recommendation

Fit a door unless CFD shows the open slot costs less than 2.4–4.1 cm².

Body shape

5.7 Decision D11

Through drag alone, body shape moves the score by hundredths of a point, since 10 cm² of drag area is worth 123 g. Mass and payload packaging decide it instead: the body must hold two container-sized units, the reel at the Mission 3 location and the sensor bay. TC-02 fixes the container size. A CAD packaging study then gives the smallest body, and any outer form (whale-like, MQ-4C-like or a plain tube) is scored on the same drag-and-mass scale (`config_exchange.pdf`).

Status

Deferred until TC-02 sets the container size.

06 Deployment and recovery

Screening

6.1 Decision D12

Eight concepts were eliminated against the rules or against physics:

- a receiver on the drum, ruled out by the §3.2.1 external switch;
- a pulley block, which needs 77–153 s to deploy against a 20 s window;
- a boom, a wireless link and a two-line bridle, all ruled out by §3.1.1;
- a spring reel;
- a clock-spring lead, which would need 23 turns;
- a sail-winch servo, which would need 2.3 times its rated torque.

Trade

6.2

The four surviving candidates were sized with datasheet drives and compared pairwise on five criteria. Each criterion is weighted by the expected points it can swing:

- 100 g of mechanism mass costs 0.046 points;
- 1 s of stow time costs 0.002 points;
- each additional 1 % chance of losing the sensor costs 0.028 points.

Reliability therefore dominates the trade.

Option A, the closed-loop drum, combines an encoder, a stop bead at the fairlead, a slip ring and a docking latch. The stop bead sets the 2.74 m deploy length and carries lap loads into the airframe. Option A is the Condorcet winner. It wins 95–98 % of draws under three of the four decision rules and 71 % under the weighted Table 06 rule (Fig. 04). The open-loop drum (B) is the runner-up: it is easier to build

and wins 29 % under the weighted table, but it ties A only if its own failure rates fall below 46 % of the estimate.

Recommendation

Option A, the closed-loop drum, at medium-high confidence.

Open evidence

6.3

The failure rates in the trade are engineering estimates. TC-05, a bench test card for the tow, reel and slip ring, must replace them with counted cycles. The model carries the verified Pololu 37D 70:1 drive (a 0.32 kg mechanism with a 12 s stow at 3.4 kg). A brushless servo, on reseller data only, would save about 0.12 kg.

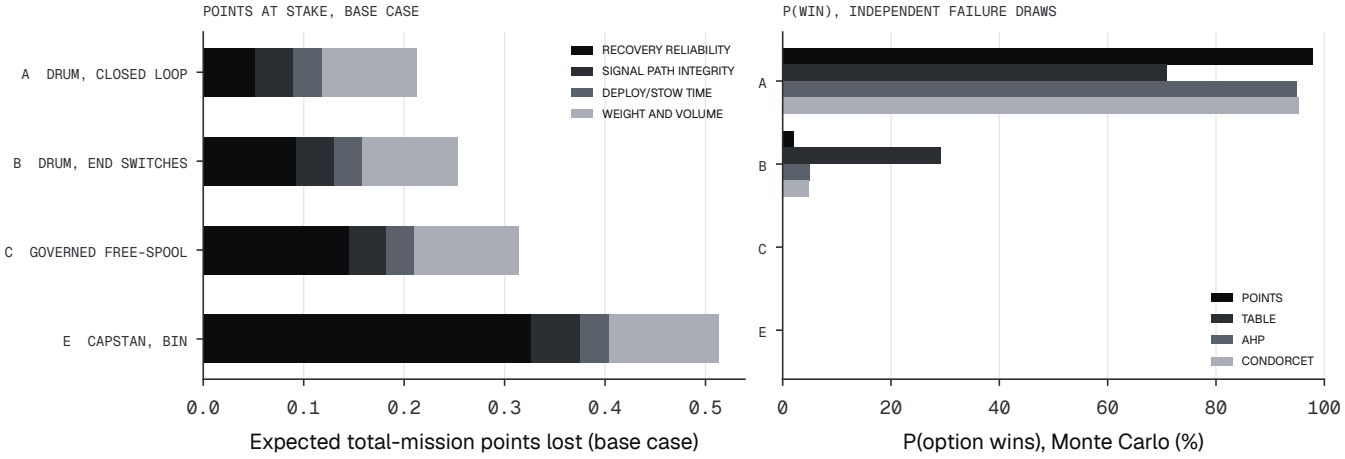


Figure 04

Recovery reliability dominates the points at stake, and the closed-loop drum (A) wins under every decision rule. Left: expected total-mission points lost by criterion, base case. Right: P(option wins) over 5,000 draws, each option's failure rates drawn independently, under four decision rules. A: closed-loop drum; B: open-loop drum; C: governed free-spool; E: capstan and bin. Source: `dr_trade_report.txt`.

07

Risk register

Table 05 lists the risks that can cost a mission, each paired with the test or analysis that retires it. Likelihood (L) and consequence (C) are graded low, medium or high.

Table 05

Risk register. Spearman values give each input's correlation with total score over the draws. Source: `s6_report.txt`.

ID	Risk	L	C	Mitigation	Retired by
R1	Airframe heavier than the fit (±31 % at 1 sd; Spearman −0.64) makes M2 infeasible	M	H	4 kg unit with optional simulators; weigh every part	CAD mass properties (WP-01 K5); first weigh-in
R2	Laps slower than modelled (median ×1.15, up to ×1.48; Spearman −0.68)	H	M	Calibrated model; margin on the M2 clock	GPS laps in flight test
R3	Stall too fast for reliable landings at full payload	M	H	14 m/s bound, flaps	Pilot sign-off; heavy-landing flight test
R4	Sensor or container fails the six GM drops (only one sensor)	M	H	Container for 1.524 m; GM after M3 (Q&A)	TC-02 with the real sensor
R5	Sensor not stowed for landing (jam, stall)	L–M	H	Closed-loop drum, stop bead, stall detection	TC-05 cycles at 3.4 kg
R6	Sensor unstable in tow (spins: M3 fails)	M	H	Fixed-geometry dart	TC-01 blower rig; CFD
R7	Lights not distinguishable in sun	M	H	Bright arrays, slow flash	TC-03
R8	Five-minute staging overrun (M3 sensor attach, M2 loading)	M	M	Tool-less attach, drilled crew	TC-04
R9	Tow line reaches the propeller or structure (SR-10)	L	H	Tractor layout; exit placed for clearance	CAD check; flight test
R10	Propulsion short of power or overheating at 100 A	M	H	Propulsion study with measured data	Thrust stand
R11	Rules read differently (GM timing, stow in window, hatches)	M	M	Conservative assumptions	Q&A round 1 (1 Nov)
R12	Parts late (PR-01 motor and prop numbers were due 14 Oct)	H	M	Order right after the propulsion study	PR-01 filed

08

Decisions requested

Table 01 serves as the decision record, and the team writes each choice into its final column.

By 17 October

D1–D4, D6–D8 and D12. D1 depends on D2, so the pilot's stall decision comes first.

By 20 October

D5, D9, D10 and D11. These wait on TC-02, the propulsion study and WP-01 CFD.

Targets enter `analysis/inputs/aircraft.toml` and the proposal only after sign-off. Table 06 of the proposal is wired once D12 is decided.

09

Sources

Analysis reports

9.1

All reports are in `analysis/out/`:

- `s4_report.txt`: sizing;
- `s6_report.txt`: Monte Carlo, robust map, tornado;
- `db01_report.txt`: stall sweep, unit table, waiver;
- `config_trades_report.txt`: exchange rate, retract break-even;
- `trades_report.txt`: matrix audit;
- `dr_trade_report.txt`: Table 06.

Field data

9.2

Field and configuration data are in `analysis/data/`. Each value carries its verbatim quote and page reference and is checked against the cached source by `verify.py`.

Primary documents

9.3

- AIAA DBF 2026–27 rules, draft of 2 September 2026 (latest official);
- AIAA competition summaries and top-3 design reports, 2021–26;
- manufacturer data for the D&R drives (Pololu, goBILDA, REV), read 13–14 October 2026.

Document control

9.4

Prepared by Badr Obtel, who is the point of contact for questions on this brief. Revision A, 14 October 2026. The brief remains provisional until the team signs the decision record in Table 01.